



Python For & While Loops

Data with Dylan | Cheat Sheet | Python Fundamentals #3

WHAT ARE LOOPS?

Loops repeat a block of code without rewriting it.

```
for i in range(5):
    print(i)
# -> 0, 1, 2, 3, 4
```

TIP: Think of loops as replay buttons: for follows a set schedule; while stops when a condition changes.

FOR LOOPS VS. WHILE LOOPS

LOOP TYPE	WHEN TO USE / LOOPS THROUGH	BASIC SYNTAX
for	Known repetitions; ranges or collections	<pre>for i in range(5): print(i)</pre>
while	Repeats while a condition is True	<pre>while i < 5: print(i)</pre>

LOOP WARNINGS

range() excludes the stop value

X `range(1, 10)` # 1-9

✓ `range(1, 11)` # 1-10

while loops must update

X `while i < 5:`
 `print(i)`

✓ `i += 1` # inside loop

MINI PRACTICE

PRACTICE: LIMITED GUESSING GAME

Build a loop that gives the player 3 attempts.

Stop early with break when the guess is correct.

```
attempts = 1
while attempts <= 3:
```

Hint: `input()` inside loop • `int()` • `attempts += 1`

KEY SYNTAX & OPERATIONS

For Loop with range()

```
for i in range(5):          # 0 through 4
    print(i)

for i in range(1, 11):     # 1 through 10
```

`range(start, stop)`: start is included; stop is excluded.

Multiplication Table + F-String

```
num = 5
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

F-strings insert variables or expressions inside {curly braces}.

Nested For Loops

```
for i in range(1, 11):
    for j in range(1, 11):
        print(f"{i} x {j} = {i * j}")
```

The inner `j` loop finishes before the outer `i` loop advances.

While Loop Structure

```
i = 0                # declare first
while i < 5:
    print(i)
    i += 1           # update inside
```

The condition is checked before every iteration.

Exit Early with break

```
if guess == secret:
    break # exit nearest loop
```

COMMON BEGINNER MISTAKES

X Wrong range limit

`range(1, 10)` # misses 10

✓ `range(1, 11)`

X Infinite while loop

```
while i < 5:
    print(i)
```

✓ `i += 1`

X Undeclared loop variable

```
while attempts <= 3:
    # placeholder
```

✓ `attempts = 3`
`while attempts <= 3:`
 `# placeholder`